

INTERNSHIP PROJECT

# Employee Leave Management System

Project Definition Document — Django + React

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|-------------|-----------------|
| Stack       | Django + React  |
| Deliverable | Working Web App |

## 1. About This Project

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You are building an Employee Leave Management System — a real product that companies use to manage staff leave requests, approvals, and balances. This is a beginner-friendly but complete full-stack project.

This document defines what you need to build. How you build it — database design, folder structure, component structure — is your responsibility. Research, think, and make decisions like a real developer.

**You are given the problem. Your job is to build the solution.**

## 2. Problem Being Solved

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Most small companies manage leaves through WhatsApp or spreadsheets. This causes:

- Employees do not know how many leave days they have left
- Managers forget to approve or reject requests
- No record of who approved what and when
- HR has no data to understand how often people are absent

Your system solves this with a simple, structured web application.

## 3. Who Uses This System

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| Role     | Who They Are         | What They Can Do                                               |
|----------|----------------------|----------------------------------------------------------------|
| Employee | Regular staff member | Apply for leave, view own history, check remaining balance     |
| Manager  | Team lead            | View and approve or reject leave requests from their team      |
| HR Admin | HR department        | Full access — manage employees, configure system, view reports |

Key rules the system must enforce:

- A manager cannot approve their own leave — it goes to HR Admin

- An employee can only see their own leave history
- A manager can only see requests from their direct team
- Only HR Admin can add leave types and public holidays

## 4. What You Must Build

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The system has 5 core modules. Build all 5. They are scoped to be achievable in 2 weeks at 4 hours per day.

### Module 1 — User Accounts & Login

Users log in with email and password. HR Admin creates all employee accounts. Each user has a role that determines what they can see and do.

- Login and logout
- Each role sees a different dashboard after login
- User profile — name, department, joining date
- HR Admin can create and deactivate employee accounts
- HR Admin can create departments and assign a manager to each

### Module 2 — Leave Types & Employee Balances

HR Admin defines the types of leave available and the number of days allowed per year. Every employee has a balance for each leave type.

- HR Admin creates leave types — e.g. Sick Leave (10 days), Casual Leave (12 days), Earned Leave (15 days)
- When a new employee is created, they automatically get a balance for each leave type
- Employees who join mid-year get a proportional balance based on their joining month
- Employee can see their balance for all leave types on their dashboard

### Module 3 — Applying & Managing Leave

Employees apply for leave through a form. The request moves through a clear set of states.

- Select leave type, start date, end date, and reason
- Option to apply for a half day
- Leave application must include a required field to nominate a 'Work Handover' or 'Proxy' employee who will handle urgent tasks.
- System counts only working days — Monday to Friday, excluding public holidays
- System checks: enough balance? No overlapping leave already approved?
- A leave request has one of these states: Pending, Approved, Rejected, Cancelled
- Employee can cancel a pending request before it is actioned
- When an approved leave is cancelled, the balance is restored

## Module 4 — Manager Approval

Managers review and action leave requests from their team.

- Manager sees a list of all pending requests from their team
- Manager approves — leave status changes to Approved, balance deducted
- Manager rejects — must write a reason, status changes to Rejected
- Employee can see the updated status and any rejection reason
- Manager's view of employees and requests is strictly limited to their direct team.
- If a manager is going on leave, they must assign someone to handle approvals while they are away

## Module 5 — HR Dashboard & Reports

HR Admin has a reporting view to understand leave activity across the company.

- Summary dashboard — total leaves this month, pending approvals, upcoming leaves this week
- Full list of all leave requests with filters — by employee, department, date range, status
- All employee balances — how many days used and remaining per person per leave type
- Export the leave list as a CSV file

## 5. Plan

Each day has a clear focus and a specific outcome. At the end of each day the listed item must be working — not just coded, but visible and functional in the browser.

| FOUNDATION & CORE FEATURES         |                                                                                                                                                                                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    |                                                                                                                                                                                                                            |
| <b>Project Setup</b>               | Django project running, PostgreSQL connected, React app running, both communicate. User model with roles created. Login API works.                                                                                         |
| <b>Auth &amp; Roles</b>            | Login page complete. JWT tokens stored correctly. After login, each role sees a different dashboard page. Logged-out users cannot access protected pages.                                                                  |
| <b>Departments &amp; Employees</b> | HR Admin can create departments and assign a manager. HR Admin can create employee accounts. Employee profile page shows name, department, joining date.                                                                   |
| <b>Leave Types &amp; Balances</b>  | HR Admin can create leave types with annual quotas. When an employee is created, balances are automatically created. Employee dashboard shows their balance for each leave type.                                           |
| <b>Apply for Leave</b>             | Employee can fill and submit a leave application form. Form validates: dates not in past, enough balance, no overlap with existing approved leave. Submitted request appears in employee's leave list with Pending status. |

| APPROVAL, REPORTING & POLISH      |                                                                                                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   |                                                                                                                                                                                         |
| <b>Manager Approval — View</b>    | Manager sees all pending requests from their team in a clear list. Each request shows employee name, leave type, dates, duration, and reason.                                           |
| <b>Manager Approval — Actions</b> | Manager can approve a request — status updates, balance deducts. Manager can reject with a written reason — status updates. Employee sees updated status immediately.                   |
| <b>Cancel &amp; Edge Cases</b>    | Employee can cancel a pending or approved request. Balance restores on cancellation. Manager's own leave goes to HR Admin for approval. All error messages are clear and user-friendly. |
| <b>HR Dashboard &amp; Reports</b> | HR Admin sees summary stats. Full filterable leave list works. All employee balance table works. Export to CSV works.                                                                   |
| <b>Testing &amp; Final Polish</b> | Test every flow end-to-end for all three roles. Fix all bugs. Loading states and empty states handled. App is clean, deployed, and demo-ready.                                          |

## 6. What You Will Learn

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| Area           | Skill                                                                        |
|----------------|------------------------------------------------------------------------------|
| Backend        | Designing a database, building REST APIs, role-based access control          |
| Business Logic | State machine for request status, balance deduction, working day calculation |
| Frontend       | Consuming a real API, managing login state, role-based page rendering        |
| Forms          | Validation on both frontend and backend, handling error responses            |
| Data           | Filtering and querying data, exporting CSV from a real dataset               |
| Process        | Git commits, environment variables, deploying a live application             |

## 7. Rules & Expectations

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### Code Quality

- Use meaningful names for variables, functions, and files — not 'data', 'stuff', or 'temp'
- Never hardcode passwords, secret keys, or API keys in your code — use environment variables
- Every user action that can fail must show a clear error message in the UI
- Commit code to Git at the end of every day with a meaningful commit message

### What Is Not Acceptable

- Copying full modules from AI or tutorials without reading and understanding every line
- Submitting code you cannot explain if asked
- Moving to Day 6 work if Day 5 is not fully working
- Skipping error handling — assume things will go wrong and handle it

## 8. Final Deliverable

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At the end of the project, you must have:

- A deployed web application accessible via a URL
- All three roles working — Employee, Manager, HR Admin
- All 5 modules functional end-to-end
- A Git repository with clean commit history
- A short demo — walk through the full flow: apply leave → manager approves → balance updates

**The goal is not to finish the project.**

*The goal is to understand every decision you make while building it.*